Electric Field and Electric Potential: Lessons From Newton's Laws

SYNOPSIS

You have heard me referring and drawing parallels to Newton's equations to get a feel for the electric field, **E**, and electric potential, *V*. The reasons are apparent. Physics phenomena are pretty distinct between gravitation and electrostatic, for sure. Still, the forms of equations are identical: If you swap out the mass for the charge and *G* for *k*, then *g* turns into **E**, and Newton's law of gravitation turns into Coulomb's law. Moreover, the weight $F_g = mg$ in Newton's equation becomes the attraction/repulse $F_E = qE$ in Coulomb's equation! Gravitational field strength (or simply gravitational field) $g = F_g/m$, the force in each kg of mass becomes electric field $\mathbf{E} = F_E/q$, the force in each C of charge. Finally, the work in the gravitational field $W_g = mgh = mU_g$ turns to the work in the electrostatic field $W_E = qEd = qV_E$, where "$gh = U_g$" is analogous to "$Ed = V_E$" which are known as gravitational potential, *U* and electrostatic potential *V*, respectively (I dropped the subscripts g and E). The gravitational potential $U = W_g/m$ is the energy in each kg of mass in a gravitational field, and the electrostatic potential $V = W_E/q$ is the energy in each C of charge in an electrostatic field.

The underlying reason for the isomorphism in equations is that both laws fundamentally seek to find the acceleration so that we can describe the motion, whether it's a mass or charge. A mass or charge will respond to its respective field, the mass in a gravitational field and the charge in an electrostatic field. Remember, the fields are "spheres of influence" created by other masses or charges whose motions are not interesting to us; only the mass or the charge that feels/experiences the field (or influence and the mass or the charge will accelerate because of the generated field) are.

As we move into the electricity topic, we will use *V*, the electrostatic potential, better known as the potential difference or voltage, with the unit volt (V, yes, the uppercase v is the symbol for electrostatic potential/potential difference/voltage and the unit of voltage). I summarize the equations in the table below.

	Newton (Mechanics)	Coulomb (Electrostatic)
Force on all bodies (1 and 2) [N]	$F_G = G M_1 M_2 / r^2$	$F_{Coulomb} = k q_1 q_2 / r^2$
Force on the object of interest (M_2 or q_2) [N]	$F_g = M_2 g$	$F_E = q_2 E$
Field [N/m] for mass (generated by M_1) [(N/C) = (V/m)] for charge (generated by q_1)	$g = G M_1 / r^2 = F_g / M_2$	$E = k q_1 / r^2 = F_E / q_2$
Potential energy (object 2) [J]	$W_g = M_2 gh = M_2 U$	$W_E = q_2 Ed = q_2 V$
Potential (object 2) [J/kg] for mass [(J/C) = V] for charge	$U = W_g / M_2 = gh$	$V = W_E / q_2 = Ed$

Finally, let me introduce a convenient energy unit for electrostatics: electronvolt, eV. The way you should think about the eV unit is (unit elementary charge) × (Voltage). Remember, voltage is the electrostatic potential or the energy per charge, so 1 electron or proton traversing 1 volt of potential has 1 eV of energy. This energy is equivalent to 1.6×10^{-19} J, i.e., numerically the same as the elementary charge (1.6×10^{-19} C).

PERFORMANCE EXPECTATION

1. **Solve** problems involving electrostatic field and electrostatic potential.

One ladder, two names

Before the questions, here is the table drawn instead of written. A **2 C charge** sitting in a uniform **3 N/C** field, beside a **2 kg mass** sitting in a uniform **9.8 m/s²** field. Climb either ladder a metre at a time and read off the potential and the work.

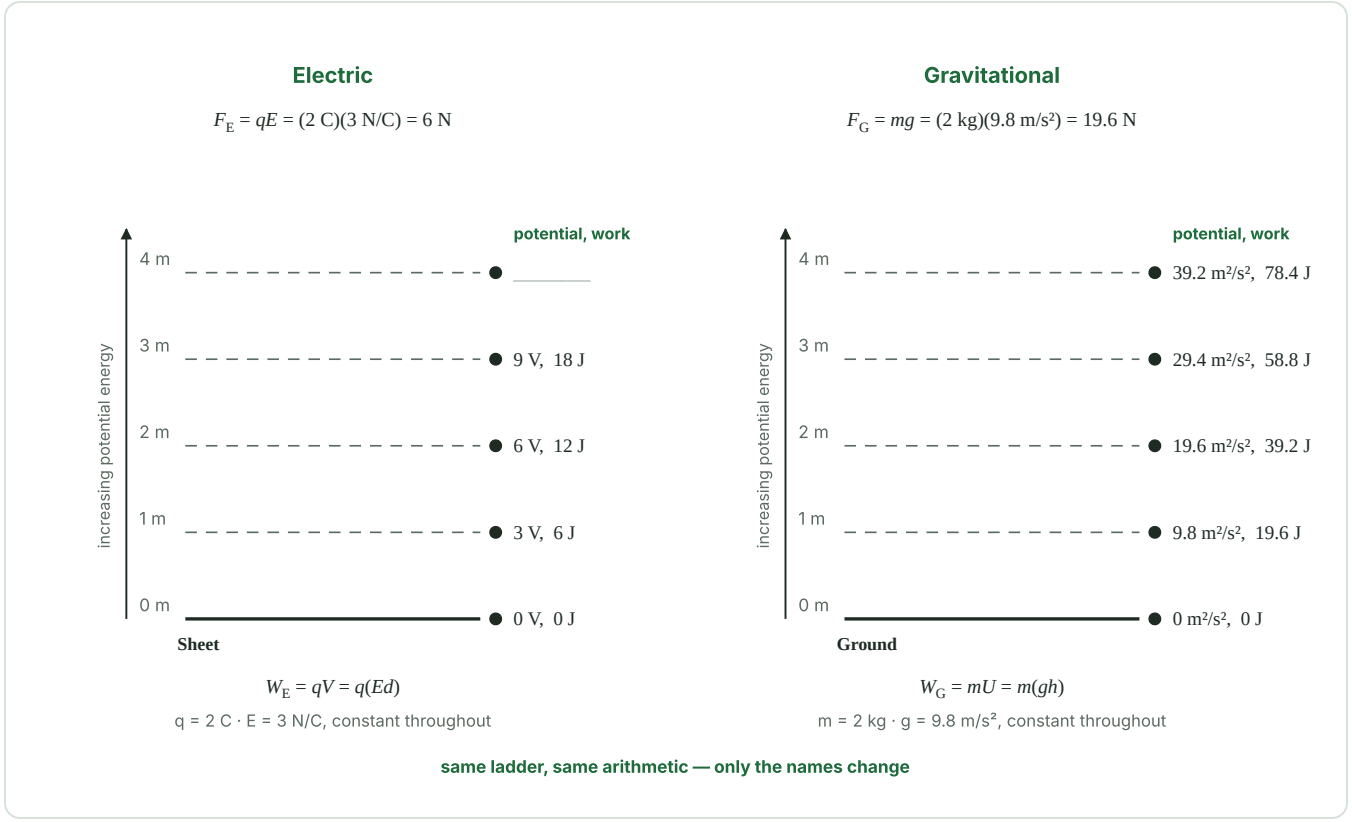


Figure 1. This is his own sheet, redrawn. Each rung is one metre. On the left the potential climbs $V = Ed$ = 3 V per metre and the work climbs $W_E = qV$; on the right it is $U = gh$ = 9.8 m²/s² per metre (that is 9.8 J/kg) and $W_G = mU$. **The 4 m rung on the electric side is blank because he left it blank** — the gravitational side is filled in all the way, so you have a worked column to copy the pattern from.

THE PUNCHLINE, IN HIS WORDS

swap the mass for the charge and G for k

“Then g turns into **E**, and Newton’s law of gravitation turns into Coulomb’s law.” Every row of the table is one line of that swap, and every question in the next two notes is one row of the table being used. If you can do the mechanics problem, you can already do the electrostatics one — you just have to recognise which row you are standing on.

Looking ahead

The table is the claim; the next two notes are the evidence. **Note 9** takes the top half of it — force and field — and works through fifteen questions with it, finishing by weighing a charge you cannot see. **Note 10** takes the bottom half, potential and work.